

(1001CJA102120114)

Test Pattern

**CLASSROOM CONTACT PROGRAMME**

(Academic Session : 2020 - 2021)

JEE(Advanced)
FULL SYLLABUS
24-07-2021
JEE(Main + Advanced) : ENTHUSIAST COURSE (SCORE-III) ALL STAR BATCH**ANSWER KEY****PAPER-1****PART-1 : PHYSICS**

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A	A,B	A,D	A,C,D	A	A	A,C	B	A,D	B	C
	Q.	11	12								
	A	D	C								
SECTION-III	Q.	1	2	3	4	5	6				
	A	1	2	2	5	2	9				

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	C	B,D	A,B,C	A,B,D	A,C,D	C,D	A,B,C	A,D	C	C
	Q.	11	12								
	A.	B	D								
SECTION-III	Q.	1	2	3	4	5	6				
	A.	6	3	5	3	6	2				

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,B,C,D	A,B,C	B,C,D	A,B,D	A,D	A,C	A,B,D	A,B,C,D	D	C
	Q.	11	12								
	A.	C	B								
SECTION-III	Q.	1	2	3	4	5	6				
	A.	5	1	8	4	6	1				

(1001CJA102120115)

Test Pattern

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24-07-2021
JEE(Main + Advanced) : ENTHUSIAST COURSE (SCORE-III) ALL STAR BATCH**ANSWER KEY****PAPER-2****PART-1 : PHYSICS**

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,C,D	A,C	B,D	A,B,D	A,B,C	A,B	D	D	A	C
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	2.82 to 2.83	4.00	12.56	1.73	1.50	32.25	2.23 to 2.25	59.00		

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,C,D	A,B,C,D	A,B,C	A,B,D	A,B,C	A,B,C,D	B	B	C	A
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	7.95 – 7.96	90.00	0.19 – 0.20	27.58 – 27.60	5.00	17.00	2.00	3.00		

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A,C	A	A,B	A,C,D	A,B,C,D	A,B	D	A	B	C
SECTION-II	Q.	1	2	3	4	5	6	7	8		
	A.	0.50	9.00	18.00	4.00	8.33	6.00	4.00	6.00		

PAPER-1**PART-1 : PHYSICS****SOLUTION****SECTION-I**

1. Ans. (A, B)

Sol. $\vec{L}_A = mV_{cm} \frac{R}{2} \hat{k} + \frac{mR^2}{2} \omega(-\hat{k}) = 0$

$$\vec{L}_B = \frac{mR^2}{2} \omega(-\hat{k})$$

$$\vec{L}_D = \frac{3}{2} mR^2 \omega(-\hat{k})$$

$$\vec{L}_O = mV_{cm} R(-\hat{k}) + \frac{mR^2}{2} \omega(-\hat{k})$$

$$= \frac{3}{2} mR^2 \omega(-\hat{k})$$

2. Ans. (A, D)

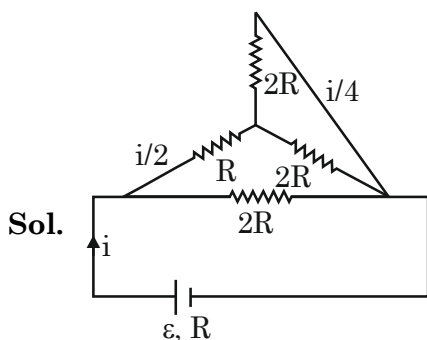
Sol. $h_0 = \frac{2S \cos \theta}{r_0 \rho g} = \frac{2 \times 7.5 \times 10^{-2}}{5 \times 10^{-4} \times 10^3 \times 10}$

$$= \frac{15 \times 10^{-2}}{5} = 3 \times 10^{-2} \text{ m} = 30 \text{ mm}$$

$$h_0 r_0 = hr$$

$$\Rightarrow r = \frac{h_0}{h} r_0 = \left(\frac{30}{25} \right) (0.5) = 0.6 \text{ mm}$$

3. Ans. (A, C, D)

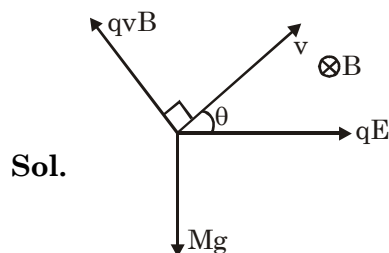


$$i = \frac{\varepsilon}{2R}$$

$$i (\text{inductor}) = \frac{\varepsilon}{8R}$$

$$Q = C V_{AB} = \frac{C\varepsilon}{2}$$

4. Ans. (A)



$$qvB \cos \theta = mg \dots\dots\dots(1)$$

$$qE = qvB \sin \theta \dots\dots\dots(2)$$

$$v \cos \theta = v_x \dots\dots\dots(3)$$

$$v \sin \theta = v_y \dots\dots\dots(4)$$

5. Ans. (A)

Sol. 2 – 3 and 4 – 1 are adiabatic processes and 1 – 2 and 3 – 4 are isothermal

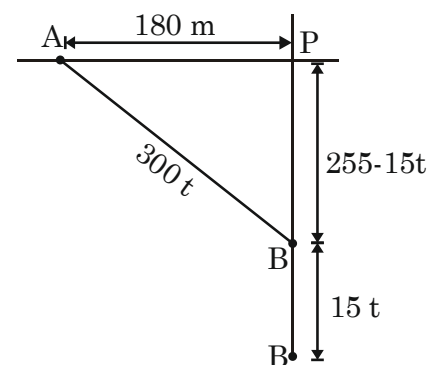
$$\therefore T_1 = T_2; T_3 = T_4$$

Work done in an adiabatic process depends on change in temperature. Change in temperature in both 2 – 3 and 4 – 1 is same.

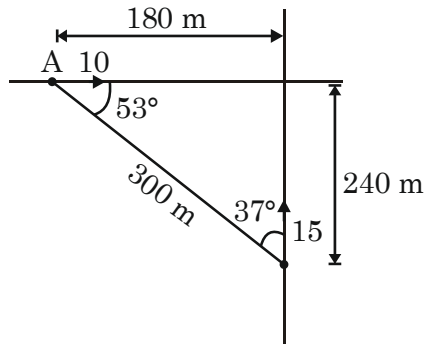
$$\therefore A_1 = A_2$$

6. Ans. (A, C)

Sol. $(180)^2 + (255 - 15t)^2 = (300t)^2$
 $32400 + 65025 + 225 t^2 - 7650 t = 90000 t^2$
 $89775 t^2 + 7650 t - 97425 = 0$
 $\Rightarrow t = 1 \text{ sec}$



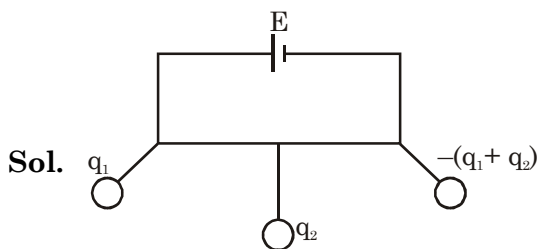
$$f = \left(\frac{300 + 15 \cos 37^\circ}{300 - 10 \cos 53^\circ} \right) f_0 = \left(\frac{300 + 12}{300 - 6} \right) f_0$$



$$= \frac{312}{294} \times 2.94 \text{ kHz}$$

$$f = 3.12 \text{ kHz}$$

7. Ans. (B)



Sol.

$$-\frac{k(q_1 + q_2)}{a} + E \Rightarrow \frac{kq_1}{a}$$

$$\frac{kq_2}{b} = \frac{kq_1}{a} - \frac{E}{2}$$

$$q_2 \Rightarrow \frac{b}{a}q_1 - \frac{bE}{2k}$$

$$-\frac{k}{a}\left(q_1 + \frac{b}{a}q_1 - \frac{bE}{2k}\right) + E = \frac{kq_1}{a}$$

$$-\frac{kq_1}{a} - \frac{kb}{a^2}q_1 + \frac{bE}{a} + E = \frac{kq_1}{a}$$

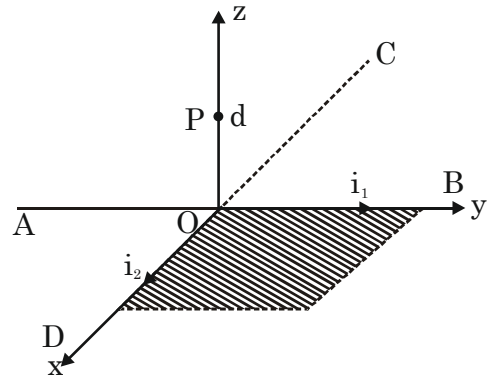
$$\frac{bE}{a} + E = \left(\frac{2k}{a} + \frac{kb}{a^2}\right)q_1$$

$$q_1 \Rightarrow \frac{aE}{2k} \Rightarrow 2\pi\epsilon_0 aE$$

$$\frac{kq_2}{b} \Rightarrow \frac{k}{a} \times \frac{aE}{2k} - \frac{E}{2} = 0$$

8. Ans. (A, D)

Sol.



$$\vec{B} \text{ due to wire (A)} \vec{B}_1 = \left(\frac{\mu_0 i_1}{2\pi d}\right) \hat{i}$$

$$\vec{B} \text{ due to wire (B)} \vec{B}_2 = \left(\frac{\mu_0 i_2}{2\pi d}\right) (-\hat{j})$$

9. Ans. (B)

Sol. By using this equation $\frac{\mu_2}{v} = \frac{\mu_1}{u} + \frac{(\mu_2 - \mu_1)}{R}$, we get all correct values of image.

$$(I) \frac{2}{v} = \frac{-1}{10} + \frac{2-1}{20}, v_1 = -40 \text{ cm}$$

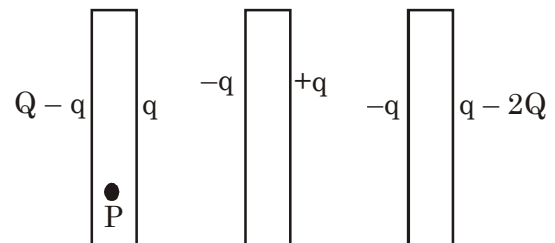
$$v_2 = \infty, v_3 = 80, v_4 = 60$$

10. Ans. (C)

Sol. By using this equation $m = \frac{v\mu_1}{u\mu_2}$, we get the correct combination.

11. Ans. (D)

Sol. Distribution of charge on different surfaces of the plates have been shown.



Take a point P on left most plate. The electric field at P

$$E = \frac{Q-q}{2A\epsilon_0} - \frac{q}{2A\epsilon_0} + \frac{q}{2A\epsilon_0} - \frac{q}{2A\epsilon_0} + \frac{q}{2A\epsilon_0} - \frac{(q-2Q)}{2A\epsilon_0} = 0$$

(inside a conductor, $E = 0$)

$$\text{So, } Q - q - q + 2Q = 0$$

Or $2q = 3Q \Rightarrow q = \frac{3Q}{2}$

The charge appearing on outer surface of right most plate

$$= q - 2Q = \frac{3Q}{2} - 2Q = -\frac{Q}{2}$$

The charge on outer surface of left most plate

$$= Q - q = Q - \frac{3Q}{2} = -\frac{Q}{2}$$

Charge appearing on middle plate is $-q$ and $+q$

i.e., $-\frac{3Q}{2}$ and $+\frac{3Q}{2}$

12. Ans.(C)

Sol. As plate is connected to earth therefore the charge on the outermost surface of left and right plate should be zero.

SECTION-III

1. Ans. 1

Sol. By using formula $N = N_0 e^{-\lambda t}$

$$\frac{dN}{dt} = a - \lambda N$$

2. Ans. 2

Sol. Current in wire $= \frac{3}{30} = 0.1$ A

$$V = IR = 2 \text{ V}$$

maximum voltage can read $= 2 \text{ V}$

$$\lambda = \frac{V}{\ell} = \frac{2}{10 \times 10^3} = 0.2 \text{ mV / mm}$$

least account of scale $= 1 \text{ mm}$

accuracy $= 0.2 \text{ mV}$.

3. Ans. 2

Sol : $\alpha = \frac{R_2 \ell_1 - R_1 \ell_2}{R_1 - R_2} = \frac{(200)(33) - (100)(67)}{100 - 200} = 1 \text{ cm}$

$$\beta = \frac{R_1 \ell_1 - R_2 \ell_2}{R_1 - R_2} - 100$$

$$= \frac{(33)(100) - (200)(67)}{100 - 200} - 100 = 1 \text{ cm}$$

4. Ans. 5

Sol. $\delta = \Sigma(\mu - 1)A$

$$= (1.5 - 1)1 + (2 - 1)2$$

$$= 0.5 + 2 = 2.5$$

5. Ans. 2

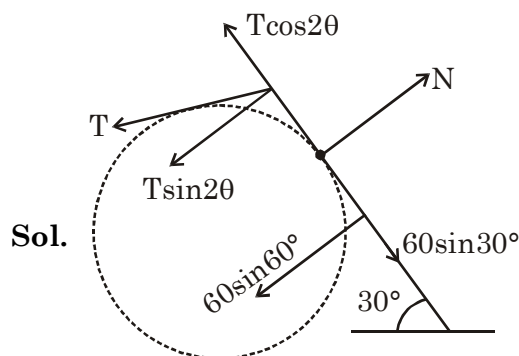
Sol. $\frac{I_{\max}}{I_{\min}} = \left(\frac{a_1 + a_2}{a_1 - a_2} \right)^2$

$$9 = \left(\frac{a_1 + a_2}{a_1 - a_2} \right)^2$$

$$\Rightarrow \frac{a_1 + a_2}{a_1 - a_2} = 3$$

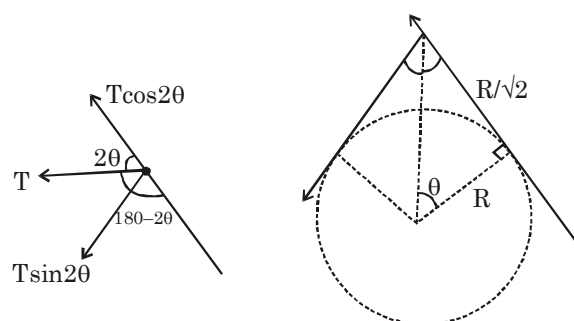
$$\Rightarrow \frac{a_1}{a_2} = \frac{2}{1}$$

6. Ans. 9



$$T \cos 2\theta = 60 \sin 30^\circ$$

$$T \sin 2\theta + 60 \cos 30^\circ = N$$



$$\tan \theta = \frac{1}{\sqrt{2}}$$

$$\cos \theta = \frac{\sqrt{2}}{\sqrt{3}}, \sin \theta = \frac{1}{\sqrt{3}}$$

$$\sin 2\theta = \frac{2}{\sqrt{3}} \times \frac{\sqrt{2}}{\sqrt{3}} = \frac{2\sqrt{2}}{3}$$

$$T \sin 2\theta \left(\frac{R}{\sqrt{2}} \right) = \tau_{mg \sin 60^\circ}$$

$$\tau_{mg \sin 60^\circ} + \tau_{mg \sin 30^\circ} = (T) R$$

$$(T \sin 2\theta) \frac{R}{\sqrt{2}} + (3g) R = TR$$

$$T - T \left(\frac{\sin 2\theta}{\sqrt{2}} \right) = 30$$

$$T(\sqrt{2} - \sin 2\theta) = 30\sqrt{2}$$

$$T = \frac{30\sqrt{2}}{\sqrt{2} - \sin 2\theta} = mg$$

$$m = \frac{3\sqrt{2}}{\sqrt{2} - \sin 2\theta}$$

$$= \frac{3\sqrt{2}}{\sqrt{2} \left[1 - \frac{2}{3} \right]} = 9\text{kg}$$

PART-2 : CHEMISTRY

SOLUTION

SECTION-I

1. Ans. (C)

2. Ans. (B, D)

Sol. (B) Due to -I effect an stabilization of conjugated base

(D) Due to -I effect second is more acidic than Ist

3. Ans. (A,B,C)

Sol. Stability of species having larger anion depend on size matching with cation
Solubility depends on low lattice energy as well as high hydration energy

4. Ans. (A, B, D)

5. Ans. (A, C, D)

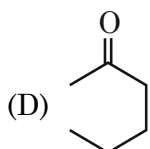
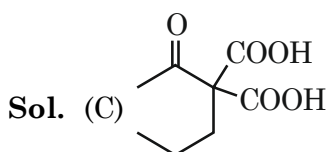
Sol. $T_b = 350 \text{ K}$

$$T_c = \frac{8a}{27Rb} = \frac{8}{27} \times 350\text{K}$$

So the gas can't be liquefied above 103.7 K

6. Ans. (C, D)

7. Ans. (A, B, C)



8. Ans. (A, D)

9. Ans. (C)

10. Ans. (C)

11. Ans. (B)

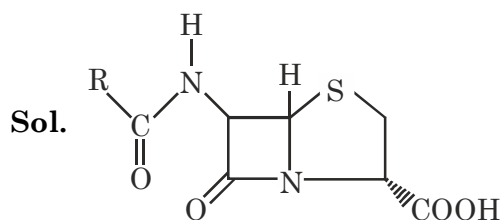
12. Ans (D)

SECTION-III

1. Ans.(6)

2. Ans.(3)

3. Ans.(5)



4. Ans.(3)

Sol. Only primary amine will give +ve test.

5. Ans.(6)

Sol. acac remains only at cis position.
Remaining combinations are
(ab) (ac) (ad) (bc) (bd) (cd) for [M (AA) abcd]

6. Ans.(2)

Sol. Coagulating value = millimole of electrolyte/volume of sol(L)
= 1 mL × 1 molar / 1/2 = 2

PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. **Ans. (A,B,C,D)**

Sol. $f'(x) = -[\sin x \cos(\cos x) + \cos x \sin(\sin x)] > 0$

$$\text{in } x \in \left[-\frac{\pi}{2}, 0\right] \Rightarrow f(x) \text{ is increasing}$$

$$f(x)_{\max} = f(0) = 1 + \sin 1, \text{ and}$$

$$f(x)_{\min} = f\left(-\frac{\pi}{2}\right) = \cos 1$$

$$f(0) \geq f(0^+) \Rightarrow 1 + \sin 1 \geq \sin a + 1$$

$$\Rightarrow \sin 1 \geq \sin a$$

$$\Rightarrow a = 1 \text{ in } \left(0, \frac{\pi}{2}\right)$$

$$\text{and } a = 1, 3 \text{ in } a \in (0, \pi)$$

2. **Ans. (A,B,C)**

Sol. mid-point SP lies on tangent at vertex

$$\Rightarrow \text{mid-point Q is } (1, 4)$$

$$\therefore \text{Equation of tangent at vertex is } x + y = 5$$

$$\text{So, } \ell(L.R.) = 4(\text{distance of } (2, -1) \text{ s from } x + y = 5)$$

$$= 8\sqrt{2}$$

3. **Ans. (B,C,D)**

Sol. equation of plane 'P' is $7x - 7y + 8z + 3 = 0$

4. **Ans. (A,B,D)**

Sol. Check L.H.D = R.H.D.

5. **Ans. (A,D)**

$$\text{Sol. } f(x) = \tan x \left[1 + \frac{2 \tan^2 x}{1 - \tan^2 x} \right] = \tan x \left[\frac{1 + \tan^2 x}{1 - \tan^2 x} \right]$$

$$= \frac{\tan x}{\cos 2x} = \frac{\sin x}{\cos x \cdot \cos 2x} = \frac{\sin(2x - x)}{\cos x \cos 2x}$$

$$\Rightarrow f(x) = \tan 2x - \tan x$$

$$\text{Now } f(1) = \tan 2 - \tan 1$$

$$f(2) = \tan 4 - \tan 2$$

.

.

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$$f(2^n) = \tan 2 \cdot 2^n - \tan 2^n$$

$$g(n) = \tan 2^{n+1} - \tan 2^n$$

6. **Ans. (A,C)**

Sol. $A = (I + B)(I - B)^{-1}$

$$A^T = [(I + B)(I - B)^{-1}]^T$$

$$= [(I - B)^T]^{-1} \cdot (I + B)^T = (I + B)^{-1} \cdot (I - B)$$

$$\therefore |A| \cdot |A^T| = 1 \therefore |A| = 1$$

$$|\text{adj}(A)| = |A|^{3-1} = 1, |2A| = 8|A| = 8$$

$$A^{-1} = ((I - B)^{-1})^{-1} \cdot (I + B)^{-1} = (I - B)(I + B)^{-1}$$

7. **Ans. (A,B,D)**

Sol. $D = -p, D_1 = -2(p - q + 6)$

$$D_2 = -2p - q + 6, \quad D_3 = 6 - q$$

for unique solution, $p \neq 0$

$$\therefore L = 20 \times 21 = 420$$

for no solution, $p = 0, q \neq 6,$

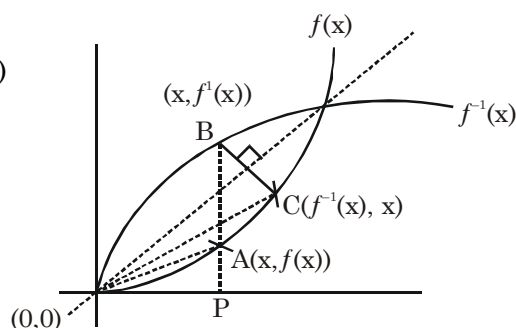
$$\therefore M = 1 \times 20 = 20$$

for infinitely many solutions

$$p = 0, q = 6, N = 1 \times 1 = 1$$

8. **Ans. (A,B,C,D)**

Sol. (A)



$$m_{OC} > m_{OA}$$

$$\frac{x}{f^{-1}(x)} > \frac{f(x)}{x}$$

$$\therefore x^2 > f^{-1}(x) f(x)$$

(B) $f'(x) = 0$ has atleast one real $\gamma \in (\alpha, \beta)$
 $\therefore f(x)f'(x) = 0$ will have roots as α, β, γ
 $\therefore$ Its derivation $(f'(x))^2 + f(x).f''(x) = 0$
 will have at least two real roots.

(C) Suppose $x = c$ ($c \neq a, b$) is a global maximum then $g'(c) = 0$ and $g''(c) \leq 0$
 But equation then say that $g''(c) = g(c)$
 $\therefore g(c) \leq 0$

Since $g(a) = g(b) = 0$, this means the maximum value of g can only be 0.

(D) Consider $g(x) = f^2(x) f(1 - 3x) (1 - 2x)$
 (apply RT)

9. Ans. (D)

10. Ans. (C)

Sol. (I) $\frac{{}^{11}C_5}{{}^{12}C_6} = \frac{1}{2}$

(II) Let E_1 be the event that S_2 and S_5 are in same group and E_2 be the event that S_2 and S_5 are in different group

$$P(E_1) = \frac{1}{11} \text{ and } P(E_2) = \frac{10}{11}$$

(using grouping)

Let E be the event that exactly one of S_2 and S_5 is among the looser then

$$P(E) = \frac{1}{11} \times 1 + \frac{10}{11} \left(\frac{1}{2} \times \frac{1}{2} + \frac{1}{2} \times \frac{1}{2} \right) = \frac{6}{11}$$

(III) Required probability = $\frac{10}{11} \times \frac{1}{2} \times \frac{1}{2} = \frac{5}{22}$

(IV) Required probability = $\frac{10}{11}$

11. Ans. (C)

12. Ans. (B)

Sol. focus = $\left(\frac{2}{3}, \frac{2}{3} \right)$

$$a = \frac{\left| \frac{2}{3} + \frac{2}{3} - 2 \right|}{\sqrt{2}} = \frac{\sqrt{2}}{3} \quad \therefore LR = \frac{4\sqrt{2}}{3}$$

Directrix is $3x + 3y - 8 = 0$

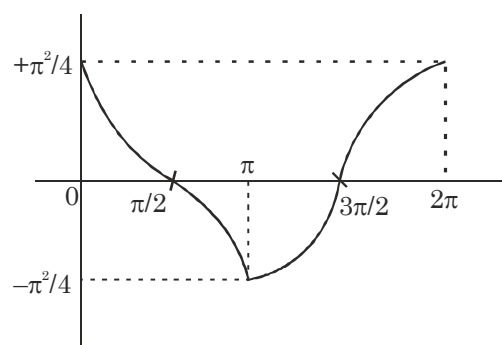
SECTION-III

1. Ans. 5

Sol. Periodic function with period = 2π

$$f(x) = \sin^{-1}(\cos x) \cdot \cos^{-1}(\sin x)$$

$$f(x) = \begin{cases} \left(\frac{\pi}{2} - x \right)^2 & , x \in \left[0, \frac{\pi}{2} \right] \\ -\left(\frac{\pi}{2} - x \right)^2 & , x \in \left(\frac{\pi}{2}, \pi \right) \\ \left(x - \frac{\pi}{2} \right) \left(x - \frac{3\pi}{2} \right) & , x \in \left[\pi, \frac{3\pi}{2} \right] \\ \left(x - \frac{3\pi}{2} \right) \left(\frac{5\pi}{2} - x \right) & , x \in \left[\frac{3\pi}{2}, 2\pi \right] \end{cases}$$



$$R_f \in \left[-\frac{\pi^2}{4}, \frac{\pi^2}{4} \right]$$

2. Ans. 1

Sol. Total number of functions = 3^{2020}

Now

$$f(1) + f(2) + f(3) + \dots + f(2020) = \text{odd integer}$$

Number of such function

$${}^{2020}C_1 \cdot 2^{2019} + {}^{2020}C_3 \cdot 2^{2017} + \dots + {}^{2020}C_{2019} \cdot 2^1$$

Now

$$(1 + 2)^{2020} = {}^{2020}C_0 \cdot 2^{2020} + {}^{2020}C_1 \cdot 2^{2019} + \dots$$

$$(2 - 1)^{2020} = {}^{2020}C_0 \cdot 2^{2020} - {}^{2020}C_1 \cdot 2^{2019} + \dots$$

$$\therefore {}^{2020}C_1 \cdot 2^{2019} + {}^{2020}C_3 \cdot 2^{2017} + \dots + {}^{2020}C_{2019} \cdot 2^1$$

$$= \frac{3^{2020} - 1}{2}$$

$\therefore$ No. of functions such that

$$f(1) + f(2) + f(3) + \dots + f(2020) = \text{even integer}$$

$$= 3^{2020} - \left(\frac{3^{2020} - 1}{2} \right) = \frac{3^{2020} + 1}{2} = N$$

$\therefore$ last digit of $N = 1$

3. Ans. 8

Sol. Divide by $\sin^2 y \cdot \cos^2 x$

$$3 \tan x \cdot \sec^2 x dx - 7 \sec^2 x dx - 7 \cot y \operatorname{cosec}^2 y dy + 5 \operatorname{cosec}^2 y dy + 4(\cot y \cdot \sec^2 x dx - \tan x \cdot \operatorname{cosec}^2 y dy) = 0$$

integrate

$$\frac{3}{2} \tan 2x - 7 \tan x + \frac{7}{2} \cot^2 y - 5 \cot y + 4(\cot y \cdot \tan x) = C$$

$$\therefore \alpha = 7, \beta = 5, \gamma = 4$$

$$\therefore \alpha + \beta - \gamma = 7 + 5 - 4 = 8$$

4. Ans. 4

$$\text{Sol. } f(x+1) = \sum_{r=0}^{200} \beta_r (x+1)^r = \sum_{r=0}^{200} \alpha_r x^r$$

$$\alpha_{200} = \beta_{200} = 1 = {}^{200}C_{200} = {}^{201}C_{201}$$

$$\alpha_{199} = \beta_{199} + {}^{200}C_{199} = {}^{199}C_{199} + {}^{200}C_{199}$$

$$= {}^{200}C_{200} + {}^{200}C_{199} = {}^{201}C_{200}$$

$$\alpha_{198} = \beta_{198} + {}^{200}C_{198} + {}^{199}C_{198} = {}^{198}C_{198} + {}^{199}C_{198} + {}^{200}C_{198} = {}^{200}C_{199}$$

⋮
⋮
⋮
⋮

$$\alpha_{100} = \beta_{100} + {}^{200}C_{100} + \dots + {}^{101}C_{100} = {}^{201}C_{101}$$

$$\begin{aligned} \therefore \sum_{r=100}^{200} \alpha_r &= \alpha_{100} + \alpha_{101} + \dots + \alpha_{200} \\ &= {}^{201}C_{101} + {}^{201}C_{102} + \dots + {}^{201}C_{201} \\ &= \frac{2^{201}}{2} = 2^{200} \end{aligned}$$

$$\therefore m = 200 \quad \therefore \frac{m}{50} = 4$$

5. Ans. 6

$$\text{Sol. Using property } \int_a^b f(x) dx = \int_a^b f(a+b-x) dx$$

and then add

$$2I = 2 \int_{-3\pi}^{3\pi} [1 + \cos x + \cos 2x + \dots + \cos 2020x] dx$$

$$2I = 2 \int_{-3\pi}^{3\pi} dx + 4 \int_0^{3\pi} [\cos x + \cos 2x + \dots + \cos 2020x] dx$$

$$I = 6\pi$$

$$\therefore K = 6$$

6. Ans. 1

$$\text{Sol. } [\bar{a} \bar{b} \bar{c}] = \bar{a} \cdot (\bar{b} \times \bar{c}) = \bar{a} \cdot [\bar{b} \times \{(\bar{a} \times \bar{c}) + \bar{b}\}]$$

$$= \bar{a} \cdot [\bar{b} \times (\bar{a} \times \bar{c})]$$

$$= \bar{a} \cdot [(\bar{b} \cdot \bar{c}) \bar{a} - (\bar{b} \cdot \bar{a}) \bar{c}]$$

$$= (\bar{a} \cdot \bar{a})(\bar{b} \cdot \bar{c}) - (\bar{a} \cdot \bar{c})(\bar{b} \cdot \bar{a})$$

$$\therefore [\bar{a} \bar{b} \bar{c}] = (\bar{b} \cdot \bar{c}) - (\bar{a} \cdot \bar{c})(\bar{a} \cdot \bar{b}) \quad \dots(i)$$

$$\bar{c} = (\bar{a} \times \bar{c}) + \bar{b}$$

$$\therefore \bar{b} \cdot \bar{c} = \bar{b} \cdot (\bar{a} \times \bar{c}) + \bar{b} \cdot \bar{b} = -[\bar{a} \bar{b} \bar{c}] + 1 \quad \dots(ii)$$

$$\bar{a} \cdot \bar{c} = \bar{a} \cdot \bar{b} \quad \dots(iii)$$

From (i), (ii) and (iii)

$$[\bar{a} \bar{b} \bar{c}] = (\bar{b} \cdot \bar{c}) - (\bar{a} \cdot \bar{c})(\bar{a} \cdot \bar{b})$$

$$= -[\bar{a} \bar{b} \bar{c}] + 1 - (\bar{a} \cdot \bar{b})^2$$

$$2[\bar{a} \bar{b} \bar{c}] = 1 - \cos^2 \theta$$

$$\therefore 2[\bar{a} \bar{b} \bar{c}]_{\max} = 1$$



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2020 - 2021)

JEE(Advanced)
FULL SYLLABUS
24-07-2021

JEE(Main + Advanced) : ENTHUSIAST COURSE (SCORE-III) ALL STAR BATCH

PAPER-2

PART-1 : PHYSICS

SOLUTION

SECTION-I

1. Ans. (A,C,D)

Sol.: $KE_{\max} = \frac{hc}{\lambda} - \phi$

$$KE_1 = \frac{1240}{200} - 4.6 = 1.6 \text{ eV}$$

$$KE_2 = 6.2 - 2.3 = 3.9 \text{ eV}$$

$$\frac{KE_1}{KE_2} = 0.41$$

$$\frac{v_{01}}{v_{02}} = \frac{\phi_1}{\phi_2} = \frac{4.6}{2.3} = 2$$

$$\frac{\lambda_{01}}{\lambda_{02}} = \frac{1}{2} = 0.5$$

Slope of Vs versus frequency

$$= \frac{h}{e} = \text{constant}$$

2. Ans. (A, C)

Sol. (A) $T = 2\pi\sqrt{\frac{\mu}{k}} = 2\pi\sqrt{\frac{m/2}{2k}} = \pi\sqrt{\frac{m}{k}}$

(B) If block is displaced by x then restoring

$$\text{force} = \frac{kx}{4}, T = 2\pi\sqrt{\frac{4m}{k}}$$

(C) If block is displaced by x, restoring

$$\text{force} = 4kx, T = 2\pi\sqrt{\frac{m}{4k}} = \pi\sqrt{\frac{m}{k}}$$

$$(D) T = 2\pi\sqrt{\frac{m}{k}}$$

3. Ans. (B,D)

Sol. $v_0 = \sqrt{\frac{GM}{a}}$

$$mv_0 a = mvr$$

$$\frac{1}{2}m\left(v_0^2 + \frac{v_0^2}{4}\right) - \frac{GMm}{a} = \frac{1}{2}mv^2 - \frac{GMm}{r}$$

$$\text{Solving, } r_{\max} = 2a, r_{\min} = \frac{2a}{3}$$

$$\text{length of semi-major axis} = \frac{4a}{3}$$

4. Ans. (A, B, D)

Sol.: For completing vertical circular motion

$$u \geq \sqrt{5g\ell}$$

$$u \geq 10 \text{ m/s}$$

$$\text{Maximum height} \leq 4 \text{ m}$$

Tension is minimum at maximum height

$$T = \left(\frac{mv^2}{\ell} - mg \right)$$

$$= \frac{m}{\ell}(u^2 - 4g\ell) - mg$$

$$= 150 \text{ N}$$

5. Ans. (A, B, C)

Sol.: $I_{\text{rms}} = \frac{E_{\text{rms}}}{Z}$

$$2.2 = \frac{220}{Z}$$

$$Z = 100 \Omega$$

$$\text{Hence } X_L = X_C$$

$$L\omega = \frac{1}{\omega C}$$

$$C = \frac{1}{\omega^2 L} = \frac{1}{(100\pi)^2 \cdot \frac{1}{\pi}} = \frac{1}{10^4 \pi} F = \frac{100}{\pi} \mu F$$

$$\therefore X_L = X_C$$

$$\text{Hence } \cos \phi = \frac{R}{C} = 1$$

$$P = E_{\text{rms}} I_{\text{rms}} = 220 \times 2.2 \times 10 = 484 \text{ W}$$

After inductor is short circuits impedance

$$Z = 100\sqrt{2}$$

$$\therefore I_{\text{rms}} = \frac{200}{100\sqrt{2}} = \left(\frac{2.2}{\sqrt{2}}\right)$$

6. Ans. (A,B)

$$\text{Sol. } \therefore \sqrt{x} = 2t - 3$$

$$x = 4t^2 - 12t + 9$$

$$v = 8t - 12$$

$$a = 8$$

$$\text{For rest } v = 0$$

$$t = 3/2 \text{ sec}$$

Distance travelled in

$$2 \text{ sec} = \int_0^2 |8t - 12| dt = 10 \text{ m}$$

7. Ans. (D)

$$\text{Sol.: } w_0 = \frac{v}{\ell}$$

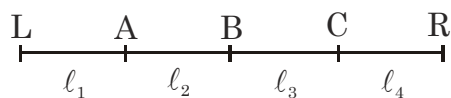
$$w_y = \frac{v}{\ell \sin \theta} = \frac{\sqrt{2} v}{\ell}$$

$$\tau_0 = mg \sin \theta \cdot \ell = \frac{mg\ell}{\sqrt{2}}$$

$$\tau_y = \frac{d\vec{L}}{dt} = 0$$

8. Ans. (D)

$$\text{Sol.: } \ell_1 + \ell_2 + \ell_3 + \ell_4 = 100$$



$$f_0 = \frac{1}{2\ell} \sqrt{\frac{T}{\mu}}$$

$$\frac{1}{\ell_1} : \frac{1}{\ell_2} : \frac{1}{\ell_3} : \frac{1}{\ell_4} = 1 : 3 : 4 : 2$$

$$\ell_1 : \ell_2 : \ell_3 : \ell_4 = 12 : 4 : 3 : 6$$

$$\therefore \ell_1 = 48 \text{ cm}$$

$$\ell_2 = 16 \text{ cm}$$

$$\ell_3 = 12 \text{ cm}$$

$$\ell_4 = 24 \text{ cm}$$

9. Ans. (A)

$$\text{Sol.: } V_c = \frac{mv + 2m \times 0 + 3m \times 0}{6m} = v/6$$

Collision between A & B

$$m \times v + 2m \times 0 = mv_1 + 2mv_2$$

$$e = \frac{v_2 - v_1}{v - 0} = 1$$

$$v_1 = -v/3 \text{ and } v_2 = \frac{2v}{3}$$

Collision between B & C

$$2m \times \left(\frac{2v}{3}\right) + 3m \times 0 = 2mv_2 + 3mv_3$$

$$e = 1 = \left(\frac{v_3 - v_2}{\frac{2v}{3} - 0} \right)$$

$$v_2 = \frac{-2v}{15} \text{ and } v_3 = \frac{8v}{15}$$

10. Ans. (C)

Sol.: $F_1 = \left(\frac{a}{4} \times \rho \times g \right) \times (a \times a/2) = \frac{1}{8} a^3 \rho g$

$$F_2 = \left(\frac{3}{4} a \rho g \right) \times (a \times a/2) = \frac{3}{8} a^3 \rho g$$

$$F_3 = \left(\frac{2}{3} a \rho g \right) \times (a \times a/2) = \frac{1}{3} a^3 \rho g$$

$$F_4 = \left(\frac{1}{3} a \rho g \right) \times (a \times a/2) = \frac{1}{6} a^3 \rho g$$

SECTION-II

1. Ans. 2.82 to 2.83

Sol.: From equation of trajectory

$$y = \frac{gx^2}{2u^2}$$

$$y_P = \frac{gx^2}{2v_P^2}$$

$$y_Q = \frac{gx^2}{2v_Q^2}$$

$$\therefore y_Q - y_P = h$$

$$g \frac{x^2}{2} \times \left(\frac{1}{v_P^2} - \frac{1}{v_Q^2} \right) = h$$

$$\begin{aligned} x &= 2\sqrt{2}h \\ &= 2 \times 1.414 \times 1 \\ &= 2.828 \\ &= 2.83 \text{ m} \end{aligned}$$

2. Ans. 4.00

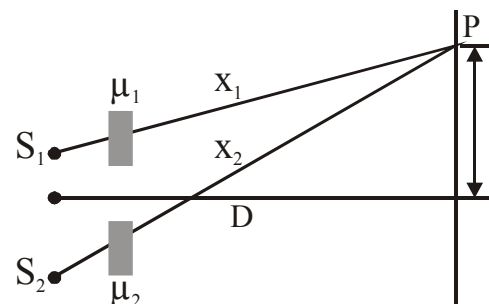
Sol. Optical path difference

$$= [(S_2P - t) + \mu_2 t] - [(S_1P - t) + \mu_1 t]$$

$$= S_2P - S_1P + (\mu_2 - \mu_1)t = \frac{Y \cdot d}{D} + (\mu_2 - \mu_1)t$$

Let P be the point where optical path difference is zero.

$$\Rightarrow \frac{Y \cdot d}{D} + (\mu_2 - \mu_1) \cdot t = 0$$



$$\Rightarrow Y = \frac{D \cdot (\mu_2 - \mu_1) \cdot t}{d}$$

But $Y = 3w$ (where w is the fringe width)

$$Y = 3 \frac{D\lambda}{d}$$

$$\Rightarrow 3 \frac{D\lambda}{d} = \frac{D}{d} (\mu_1 - \mu_2) \cdot t$$

$$t = \frac{3\lambda}{(\mu_1 - \mu_2)} = \frac{3 \times 4000 \times 10^{-10}}{(1.7 - 1.4)} = 4 \mu\text{m}$$

3. Ans. 12.56

Sol.: From Ampere's law

$$\oint \vec{B} \cdot d\vec{l} = \mu_0 I_{in}$$

$$B 2\pi r = \mu_0 \int_0^r \frac{J_0 r}{R} 2\pi r dr$$

$$\begin{aligned} B &= \frac{\mu_0 J_0}{3R} r^2 \\ &= \frac{4\pi \times 10^{-7} \times 10^4 \times (6 \times 10^{-3})^2}{3 \times 12 \times 10^{-3}} \\ &= 12.56 \times 10^{-6} \text{ T} \end{aligned}$$

4. Ans. 1.73

Sol.: $\varepsilon = \frac{B\omega L^2}{2}$

Loss in potential energy = gain in kinetic energy

$$mg \frac{L}{2} = \frac{1}{2} \times \frac{mL^2}{3} \times \omega^2$$

$$w = \sqrt{\frac{3g}{L}}$$

$$\therefore \varepsilon = \frac{B}{2} \sqrt{\frac{3g}{L}} \cdot L^2$$

$$= \frac{20}{2} \times \sqrt{\frac{3 \times 10}{0.1}} \times (.1)^2$$

$$= 1.73 \text{ volt}$$

5. Ans. 1.50

Sol. From conservation of energy

$$\frac{1}{2}mv_0^2 - \frac{GMm}{R} = \frac{1}{2}mv^2 - \frac{GMm}{R+h}$$

$$(V_0^2 - V^2) = 2 \frac{GMh}{R(R+h)}$$

$$V_1^2 - V^2 = \frac{4GM}{3R} \dots\dots\dots(1)$$

From conservation of angular momentum about centre of earth

$$mv_0R = mv(R+h)$$

$$v = \frac{v_0R}{R+h} = \frac{v_0}{3} \dots\dots\dots(2)$$

$$\therefore v_0^2 - \frac{v_0^2}{9} = \frac{4}{3} \frac{GM}{R}$$

$$\frac{8}{9}n.g.R = \frac{4}{3}.gr$$

$$n = \frac{3}{2}$$

$$n = 1.50$$

6. Ans. 32.25

Sol.: 1MSD = 1mm

$$1 \text{ VSD} = \frac{15}{16} \text{ mm}$$

$$\text{L.C.} = 1 - \frac{15}{16} = \frac{1}{16} \text{ mm}$$

$$\therefore \text{Reading} = 32 \times 1 + 4 \times \frac{1}{16} = 32.25 \text{ mm}$$

7. Ans. 2.23 to 2.25

Sol. $P = t$

$$\frac{dk}{dt} = t$$

$$K = \frac{t^2}{2}$$

$$\frac{1}{2} \times 1 \times v^2 = \frac{t^2}{2}$$

$$v = t$$

$$\therefore a_t = \frac{dv}{dt} = 1 \text{ m/s}^2$$

$$a_c = \frac{v^2}{R} = \frac{2^2}{2} = 2$$

$$\therefore a_{\text{net}} = \sqrt{a_t^2 + a_c^2} = \sqrt{5} = 2.24 \text{ m/s}^2$$

8. Ans. 59.00

Sol. In steady state

$$912 \times a\pi r^2 = \sigma \times e \times 4\pi r^2 (T^4 - T_s^4)$$

$$\frac{912}{5.7 \times 10^{-8} \times 4} = T^4 - T_s^4$$

$$\begin{aligned} T^4 &= 40 \times 10^8 + (300)^4 \\ &= 40 \times 10^8 + 81 \times 10^8 \\ &= 121 \times 10^8 \end{aligned}$$

$$T^2 = 11 \times 10^4$$

$$\begin{aligned} T &= \sqrt{11} \times 10^2 \\ &= 332 \text{ K} = 59^\circ \text{C} \end{aligned}$$

PART-2 : CHEMISTRY

SOLUTION

SECTION-I

1. Ans. (A, C, D)

Sol. Aniline + HCl → Anilinium chloride

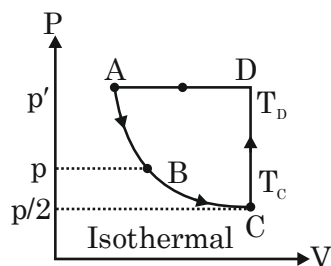
$$\begin{array}{ccc} 5 & 2.5 & 0 \\ 2.5 & & 2.5 \\ 6 = \text{Pkb} \end{array}$$

$$\text{pH} = \frac{1}{2} [14 - 6 - \log 10^{-2}] = 5$$

$$\text{pOH} = 6 + \log \frac{2.5}{7.5} = 5.52$$

2. Ans. (A, B, C, D)

Sol.



$$\Delta s_{\text{CD}} = 4 \ln 16 = nC_v \ln \frac{T_D}{T_C}$$

$$\frac{T_D}{T_C} = 16 \frac{p'}{p/2}$$

$$p' = 8p$$

3. Ans. (A,B,C)

4. Ans. (A, B, D)

5. Ans. (A,B,C)

Sol. Acidic strength of oxyacids depend on oxidation number of central atom as well as extent of resonance in canonical structures of conjugate base.

6. Ans. (A,B,C,D)

7. Ans. (B)

8. Ans. (B)

9. Ans. (C)

10. Ans. (A)

SECTION-II

1. Ans.(7.95 – 7.96)

$$\text{Sol. } \Delta x \times m\Delta V \geq \frac{h}{4\pi}$$

$$\Rightarrow \lambda \times m\Delta V \geq \frac{h}{4\pi}$$

$$\Rightarrow \frac{h}{mv} \times m\Delta v \geq \frac{h}{4\pi}$$

$$\Rightarrow \frac{\Delta v}{v} \geq \frac{1}{4\pi}$$

$$\Rightarrow \frac{h}{mv} \times 100 \geq \frac{1}{4\pi} \times 100 \approx 8$$

2 Ans.(90.00)

Sol. $A_4(g) \rightarrow 4A(g)$

$$\begin{array}{ccc} a & & 0 \\ a - x & & 4x \\ \text{Given } a - x = 4x \end{array}$$

$$\text{or } x = \frac{a}{b}$$

$$t_{\frac{1}{2}} = \frac{0.693}{K} \approx 90 \text{ min}$$

3. Ans.0.19 – 0.20

Sol. Let degree of dissociation of sucrose be α .

$\Rightarrow$ Sucrose + water $\rightarrow$ glucose + fructose

$$i = 1 + (2 - 1)\alpha = 1 + \alpha$$

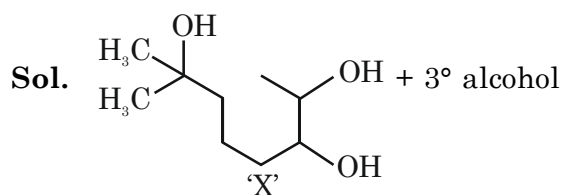
$$\Rightarrow \Delta t_b = iK_n m$$

$$\Rightarrow 0.7 = (1 + \alpha) \times 1.8 \times \frac{10 \times 1000}{342 \times 90}$$

$$\Rightarrow 1 + \alpha = \frac{0.7 \times 342 \times 9}{1.8 \times 1000} = 1.2$$

$$\Rightarrow \alpha = 0.2$$

4. Ans. 27.58 – 27.60

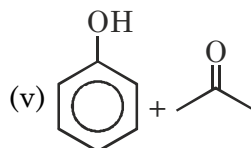
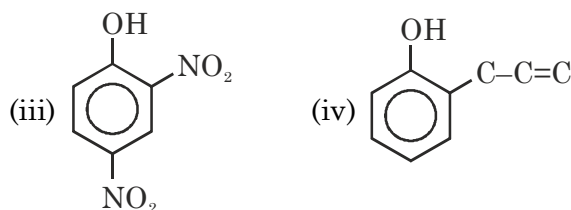
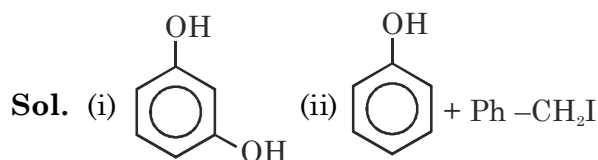


Molecular mass of the final major product is = 174

Now mass percentage of oxygen

$$= \frac{48}{174} \times 100 = 27.586\%$$

5. Ans. 5.00



6. Ans. (17.00)

[Ni(CO)₄], SO₂, SO₂Cl₂

7. Ans. 2.00

Sol. (Rb, Cs)

8. Ans. 3.00

Sol. Cold and dilute NaOH, Na₂S₂O₃, Na metal

PART-3 : MATHEMATICS

SOLUTION

SECTION-I

1. Ans. (A,C)

Sol. (A) For f(x) to be invertible f(x) should be

one-one that is $f'(x) \geq 0 \Rightarrow a \in \left[\frac{1}{4}, \infty\right)$

(B) $f(x + \pi) = \cos(\sin(x + \pi)) + \cos(\cos(x + \pi))$
 $\Rightarrow f(x + \pi) = \cos(\sin x) + \cos(\cos x) = f(x)$

$$f\left(x + \frac{\pi}{2}\right)$$

$$= \cos\left(\sin\left(x + \frac{\pi}{2}\right)\right) + \cos\left(\cos\left(x + \frac{\pi}{2}\right)\right)$$

$$= \cos(\cos x) + \cos(\sin x) = f(x)$$

$$\text{fundamental period} = \frac{\pi}{2}$$

$$(C) \frac{x^3 + 1}{2} = 2\sqrt[3]{2x - 1}$$

$$\text{Let } f(x) = \frac{x^3 + 1}{2} \Rightarrow f^{-1}(x) = \sqrt[3]{2x - 1}$$

Equation becomes $f(x) = f^{-1}(x)$

$$\Rightarrow f(x) = x \Rightarrow \frac{x^3 + 1}{2} = x \Rightarrow x^3 - 2x + 1 = 0$$

$$\Rightarrow (x - 1)(x^2 + x - 1) = 0 \Rightarrow x = 1, \frac{-1 \pm \sqrt{5}}{2}$$

2. Ans. (A)

$$\text{Sol.: } \sin \frac{\theta}{2} = \frac{r_2 - r_1}{r_2 + r_1} \Rightarrow \frac{r_1}{r_2} = \frac{1}{3}$$

$$\therefore r_2 = 3^2, r_3 = 3^3$$

$$\therefore x_1 x_2 x_3 \dots \infty = \text{cis}\left(\frac{\pi}{3}\right) \cdot \text{cis}\left(\frac{\pi}{3^2}\right) \dots \infty$$

$$= \text{cis}\left(\frac{\pi}{3} + \frac{\pi}{3^2} + \dots \infty\right) = \text{cis}\left(\frac{\pi}{3}\left(1 + \frac{1}{3} + \frac{1}{3^2} + \dots \infty\right)\right) = i$$

3. Ans. (A,B)

Sol. $\therefore A$ is an orthogonal matrix

$$\Rightarrow |A| = \pm 1 \text{ and } |\text{adj}(\text{adj} A)| = |A|^{(n-1)^2} = \pm 1$$

4. Ans. (A,C,D)

Sol. $R = 8r$

$$(\text{Apply } r = 4R \sin \frac{A}{2} \sin \frac{B}{2} \sin \frac{C}{2})$$

$$\Rightarrow \left(\sin \frac{A}{2} - \frac{1}{4}\right)^2 = 0$$

$$\Rightarrow \sin \frac{A}{2} = \frac{1}{4}$$

$$\therefore \frac{b}{c} = \frac{3 + \sqrt{5}}{2}$$

5. Ans. (A,B,C,D)

Sol.: $A \rightarrow |z| = |z-1| \Rightarrow z\bar{z} = (z-1)(\bar{z}-1)$
 $\Rightarrow z + \bar{z} = 1$ (Not possible)
 $|z| = |z+1| \Rightarrow z + \bar{z} = -1$ (Not possible)

No such z is possible

$B \rightarrow \left(\frac{z}{|z|}\right)^2 + \frac{z}{|z|} + 1 = 0$

$\Rightarrow \frac{z}{|z|} = w, w^2$

$\Rightarrow z = |z|w, |z|w^2$ (infinitely many z)

$C \rightarrow \max |z| = \sqrt{3} + 1$

$D \rightarrow z = i, 1-i, -1-i$

$\Delta ABC = 2 \text{ sq. units}$

6. Ans. (A,B)

Sol.: Given limit $= e^{\left(\frac{(\sin \alpha)^x + (\sin \beta)^x - 2}{2}\right)^{\frac{1}{x}}} = e^{\frac{\ln \sin \alpha + \ln \sin \beta}{2}}$

$= \sqrt{\sin \alpha \sin \beta} = 1$ this is possible only

when $\alpha = \beta = \frac{\pi}{2}$

7. Ans. (D)

Sol. (P) $\because x, y$ are integers
 so the solutions are $(0, 0), (1, 0),$
 $(0, 1), (-1, 0), (0, -1), (1, 1), (1, -1),$
 $(-1, 1), (-1, -1)$

(Q) Period of $f(x) = |\sin 2x| - |\sin 3x|$ is π .
 In $[0, \pi]$ $f(x) = 0$ is having 6 solutions.

i.e. $0, \frac{\pi}{5}, \frac{2\pi}{5}, \frac{3\pi}{5}, \frac{4\pi}{5}, \pi$

So, total number of solutions
 $= 4 \times 7 + 4 \times 7 + 15$

(R) $[\cos x] + [\sin x] = -1$
 $\Rightarrow [\cos x] = -1$ and $[\sin x] = 0$ or $[\cos x]$
 $= 0$ and $[\sin x] = -1$

$\Rightarrow x \in \left(\frac{\pi}{2}, \pi\right] \cup \left[\frac{3\pi}{2}, 2\pi\right) \Rightarrow \tan x \in (-\infty, 0]$

(S) $[\sin x] + [\sqrt{2} \cos x] = -3$

$\Rightarrow -1 \leq \sin x < 0 \Rightarrow x \in \left(\pi, \frac{3\pi}{2}\right]$

$\Rightarrow x \in \left(\pi, \frac{5\pi}{4}\right] \dots(1)$

$\Rightarrow -2 \leq \sqrt{2} \cos x < -1 \Rightarrow x \in \left[\pi, \frac{5\pi}{4}\right)$

$\Rightarrow x \in \left(\pi, \frac{5\pi}{4}\right] \dots(2)$

From (1) and (2) $x \in \left(\pi, \frac{5\pi}{4}\right)$

$\Rightarrow 2x \in \left(2\pi, \frac{5\pi}{2}\right)$

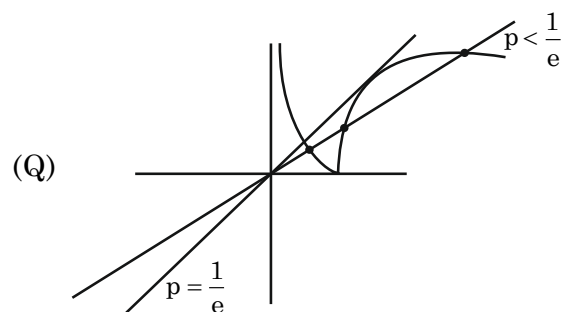
$\therefore [\sin 2x] = 0$

8. Ans. (A)

Sol.: (P) $2(x+y) = 36$

$V = \pi x(18-x)^2$

$V_{\max} \Rightarrow x = 6, y = 12$



(R) $f'(x) = ax^2 + 2(a+2)x + (a-1)$
 Both roots must be negative

(i) $D > 0 \Rightarrow a > \frac{-4}{5}$

(ii) $-\frac{b}{2a} < 0 \Rightarrow a \in (0, \infty)$

(iii) $a f(0) > 0 \Rightarrow a > 1$
 $\therefore a \in (1, \infty)$

(S) $f''(x) = 12x^2 + 6ax + 3 > 0 \forall x \in \mathbb{R}$

$\Rightarrow 36a^2 - 12(12) < 0$

$\Rightarrow a^2 - 4 < 0$

$\Rightarrow -2 < a < 2$

9. Ans. (B)

Sol. (P) Let $l = 2r + 1, m = 2r + 3, n = 2r + 5$
 where r is an integer

$\therefore (2r+1)x - (2r+3)y + (2r+5) = 0$
 $2r(x-y+1) + (x-3y+5) = 0$

It passes through the intersection of
 $x-y+1=0$ and $x-3y+5=0$

Which is $(1, 2)$

- (Q) The equation to the tangent at (2, 4) to $y = x^2$, $4x - y - 4 = 0$
The normal at (2, 4) $x + 4y = 18$
It passes through the centre of the circle which is (h, k)
 $\therefore h + 4k = 18$
Also $(h - 0)^2 + (k - 2)^2 = (h - 2)^2 + (k - 4)^2$
 $\Rightarrow h + k = 4$

Solving, we get $\left(-\frac{2}{3}, \frac{14}{3}\right)$

- (R) For the conormal point

$$y_1 + y_2 + y_3 = 0$$

$$2 + 4 + y_3 = 0 \quad y_3 = -6$$

$\therefore$ 3rd conormal point is (9, -6)

- (S) The ellipse is $4x^2 + 9y^2 = 75$

$$\text{The line is } 2x + 3y = 12$$

$$\text{Solving } x^2 - 6x + 9 = 0$$

Equal roots ($\because$ tangent)

$$x = 3$$

$$y = 2 \text{ points is } (3, 2)$$

10. Ans. (C)

Sol.: (P) Let A_i ($i = 1, 2, 3, 4$) be the event that urn contains 2, 3, 4 or 5W balls and B the event that two white balls is drawn

we have to find $P\left(\frac{A_4}{B}\right)$

Applying, Bayes theorem

$$P\left(\frac{A_4}{B}\right) = \frac{\frac{1}{4} \times 1}{\frac{1}{4} \times \frac{1}{10} + \frac{1}{4} \times \frac{3}{10} + \frac{1}{4} \times \frac{6}{10} + \frac{1}{4} \times 1} = \frac{1}{2}$$

- (Q) $n, n + 3, \dots, n + 12$

$$n + 1, \dots$$

$$n + 2, \dots, n + 14$$

Required probability

$$= \frac{3 \times {}^5C_3 + 5^3}{{}^{15}C_3} = \frac{31}{91}$$

- (R) Required probability $= \frac{3!(2)}{4 \times 4 \times 4} = \frac{3}{16}$

- (S) For the balls in box there are three possibility

E_1 : All the 4 balls are red

E_2 : 3 or the 4 balls are red

E_3 : 2 of the 4 balls are red

Assuming that E_1, E_2 and E_3 are given to be equally likely

$$P(E_1) = P(E_2) = P(E_3) = \frac{1}{3}$$

Let A be the event two balls drawn are red

$$P(A) = \sum_{i=1}^3 P(E_i)P\left(\frac{A}{E_i}\right) = \frac{1}{3} \times 1 + \frac{1}{3} \times \frac{1}{2} + \frac{1}{3} \times \frac{1}{6} = \frac{5}{9}$$

Let B is the event that next draw is red ball.

$$P\left(\frac{B}{A}\right) = \frac{P(B \cap A)}{P(A)} = \frac{\frac{1}{3} \times \frac{{}^2C_2 \times 2}{{}^4C_2} + \frac{1}{3} \times \frac{{}^3C_2 \times 3}{{}^4C_2} + \frac{1}{3} \times \frac{{}^4C_2 \times 4}{{}^4C_2}}{\frac{5}{9}} = \frac{7}{8}$$

SECTION-II

1. Ans. 0.50

Sol. In LHS Put $x + \frac{\pi}{4} = \frac{\pi}{2} + t$

$$I = \frac{1}{\sqrt{2}} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{\sec t}{e^t + 1} dt$$

$$I = \frac{1}{\sqrt{2}} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{e^t \sec t}{e^t + 1} dt$$

$$2I = \frac{1}{\sqrt{2}} \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sec t dt$$

$$\therefore \lambda = \frac{1}{2\sqrt{2}}$$

2. Ans. 9.00

Sol.: Using the idea of the differentiation of determinant, we get

$$f'(x) = \begin{vmatrix} \cos(x + \alpha) & \cos(x + \beta) & \cos(x + \gamma) \\ \cos(x + \alpha) & \cos(x + \beta) & \cos(x + \gamma) \\ \cos(\beta - \gamma) & \cos(\gamma - \alpha) & \cos(\alpha - \beta) \end{vmatrix}$$

$$+ \begin{vmatrix} \sin(x + \alpha) & \sin(x + \beta) & \sin(x + \gamma) \\ -\sin(x + \alpha) & -\sin(x + \beta) & -\sin(x + \gamma) \\ \cos(\beta - \gamma) & \cos(\gamma - \alpha) & \cos(\alpha - \beta) \end{vmatrix}$$

$$+ \begin{vmatrix} \sin(x+\alpha) & \sin(x+\beta) & \sin(x+\gamma) \\ \cos(x+\alpha) & \cos(x+\beta) & \cos(x+\gamma) \\ 0 & 0 & 0 \end{vmatrix}$$

$$= 0 + 0 + 0$$

$$f'(x) = 0 \text{ for all } x \in \mathbb{R}$$

$$f(x) = \text{constant}$$

$$\therefore f(x) = \lambda \text{ for all } x \in \mathbb{R}$$

$$\frac{\sum_{k=1}^9 f(k)}{\lambda} = 9$$

3. Ans. 18.00

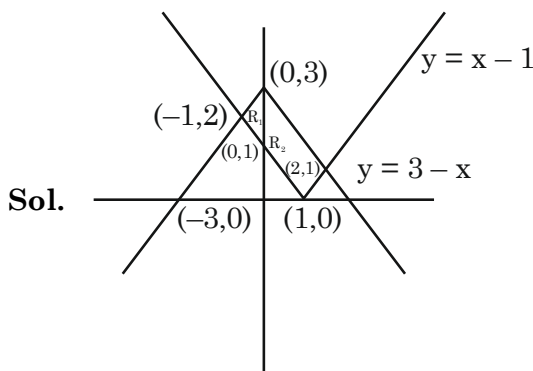
Sol. Total – (when 2 a's are in same row / column and b's are in different + when 2b's are in same row / column and a's are in different. + when when 2a's and 2b's are in same row or column.

$$= {}^{16}C_4 \times \frac{4!}{2!2!} - \left[({}^4C_1 \times {}^4C_2 ({}^{14}C_2 - 1 - {}^3C_1 \times {}^4C_2) \times 2) \times 2 \right] + {}^4C_1 \times \frac{4!}{2!2!} \times 2$$

$$= 3960$$

$$\therefore \text{sum of digits} = 18$$

4. Ans. 4.00



$$R_1 = \frac{1}{2} \times 1 \times 2 = 1$$

$$R_2 = \frac{1}{2} \times 2 \times 2 + \frac{1}{2} \times 2 \times 1 = 3$$

5. Ans. 8.33

$$\text{Sol. } \therefore f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{f(x) + f(h) + 2x \cdot h(x+h) - \frac{1}{3} - \left(f(x) + f(0) - \frac{1}{3} \right)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h} + 2x^2$$

$$= f'(0) + 2x^2 \Rightarrow f(x) = f'(0)x + \frac{2x^3}{3} + C$$

$$f(x) = \frac{2x^3 + 4x + 1}{3}$$

$$\therefore f(2) = \frac{25}{3}$$

6. Ans. 6.00

Sol.: Unit vectors along AB, AD, AC are

$$\frac{\hat{i} - \hat{j}}{\sqrt{2}}, \frac{\hat{i} - \hat{j} - \hat{k}}{\sqrt{3}}, \frac{-\hat{i} + \hat{k}}{\sqrt{2}}$$

$$\left(\begin{array}{ccc} \hat{i} & \hat{j} & \hat{k} \\ \text{vector along AB} & = & \begin{vmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \end{vmatrix} = \hat{i} - \hat{j} \end{array} \right)$$

$$\overline{AB} = \hat{i} - \hat{j}, \overline{AD} = \hat{i} - \hat{j} - \hat{k}, \overline{AC} = -\hat{i} + \hat{k}$$

$$\therefore \text{Volume} = \left| \frac{1}{6} \begin{vmatrix} 1 & -1 & 0 \\ 1 & -1 & -1 \\ -1 & 0 & 1 \end{vmatrix} \right| = \frac{1}{6}$$

7. Ans. 4.00

Sol.: Tangent equation in $\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} = 1 \rightarrow (1)$

Auxiliary circle is $x^2 + y^2 = a^2$

Homogenizing

$$\text{we get } x^2 + y^2 - a^2 \left(\frac{x \cos \theta}{a} + \frac{y \sin \theta}{b} \right)^2 = 0$$

These lines are perpendicular

$$\Rightarrow a^2 \left(\frac{\cos^2 \theta}{a^2} + \frac{\sin^2 \theta}{b^2} \right) = 2 \Rightarrow \cos^2 \theta + \frac{a^2}{b^2} \sin^2 \theta = 2$$

$$\frac{a^2}{b^2} \sin^2 \theta - \sin^2 \theta = 1 \Rightarrow \frac{a^2}{b^2} = \frac{1 + \sin^2 \theta}{\sin^2 \theta}$$

$$\therefore e = \sqrt{1 - \frac{b^2}{a^2}} = \sqrt{1 - \left(\frac{\sin^2 \theta}{1 + \sin^2 \theta} \right)} = \sqrt{\frac{1}{1 + \sin^2 \theta}}$$

$$\therefore (\alpha + \beta)^2 = 4$$

8. Ans. 6.00

$$\text{Sol.: } |(\alpha + \beta)(\alpha - \beta)| = \frac{7}{4}$$

$$\text{i.e., } \frac{7}{2} \sqrt{\frac{49}{4} - 2C} = \frac{7}{4} \Rightarrow C = 6$$